

Student Case Study – Git & GitHub (Python / Java / HTML Beginners)

Scenario

You have joined TechNova Solutions as a Graduate Associate. Your team has been asked to build a simple application using only the technologies you have learned so far (Python *or* Java *or* HTML). Your objective is to use Git and GitHub to manage your project and collaborate with your team.

Instructions

Choose **ONE** of the following projects:

- Python – Student Marks Calculator
- Java – Student Management System (Console Application)
- HTML – Personal Portfolio Website

Answer the following by performing the tasks on your computer.

1. Create a project folder for your chosen application and initialize it as a Git repository.
2. Create the initial project files (for example: main.py, Main.java, or index.html).
3. Check the repository status. What does Git display for the newly created files?
4. Stage all files and create your first commit with an appropriate commit message.
5. Modify your program by adding one new feature (for example: grade calculation, student search, or an About Me section).
6. Display the changes made before committing them.
7. Create a new branch called **feature-update** and switch to it.
8. Complete the new feature and commit the changes.
9. Switch back to the main branch. Is the new feature visible? Record your observation.
10. Merge the feature-update branch into the main branch.
11. Create a GitHub repository with the same project name.
12. Connect your local repository to GitHub.
13. Push your project to GitHub.
14. Create another branch called **bug-fix** and make a small improvement (fix spelling, improve output, or add comments).
15. Push the bug-fix branch and create a Pull Request on GitHub.
16. Merge the Pull Request into the main branch.
17. Display the complete commit history in graph format.
18. Explain the difference between Git and GitHub in your own words.
19. Explain why developers use branches instead of working directly on the main branch.
20. Submit your GitHub repository URL and screenshots of git status, commit history, branches, Pull Request, and final repository.

Deliverables

- GitHub Repository URL
- Screenshots of important Git commands
- Final project files
- Reflection: What did you learn about Git and GitHub?